

CPEN 416 / EECE 5710

Lecture 18: Quantum channels; noise in quantum systems

Thursday 13 November 2025

Announcements

- Quiz 8 on Tuesday (L16-L18)
- Monday's tutorial: observables and expectation values
- Final week schedule updates
 - Mon. 1 Dec: last tutorial, exam review
 - Tues. 2 Dec: quiz 10, groups 01-02 presentations
 - Thurs. 4 Dec: groups 03-05 presentations

Last time

$$\sum_i p_i = 1$$

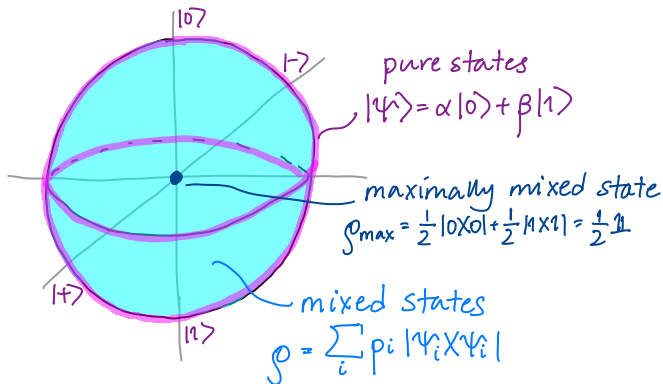
	Pure state	Pure state ρ	Mixed state ρ
States	$ \psi\rangle$	$\rho = \psi\rangle\langle\psi $	$\rho = \sum_i p_i \psi_i\rangle\langle\psi_i $
Ops.	$ \psi'\rangle = U \psi\rangle$	$\rho' = U\rho U^\dagger$ $ \psi'\rangle\langle\psi' = U \psi\rangle\langle\psi U^\dagger$	$\rho' = U\rho U^\dagger$
Meas.*	$ \langle\varphi_i \psi\rangle ^2$ $\langle\psi B \psi\rangle$	$\text{Tr}(\varphi_i\rangle\langle\varphi_i \psi\rangle\langle\psi)$ $\text{Tr}(B\rho)$	$\text{Tr}(P_i \rho)$ $\text{Tr}(B\rho) = \langle B \rangle$

* where $\{\varphi_i\}$ form an orthonormal basis, and $\{P_i\}$ is a set of projectors $P_i = |\varphi_i\rangle\langle\varphi_i|$ or more generally a POVM ($\sum_i P_i = I$).

Last time

$$\langle X \rangle = \text{Tr}(X\rho) \quad \langle Y \rangle = \text{Tr}(Y\rho) \quad \langle Z \rangle = \text{Tr}(Z\rho)$$

(See L17 notes for derivation)



Learning outcomes

- 1 Define and apply quantum channels to quantum states
- 2 Express noise in quantum systems using quantum channels
- 3 Compare density matrices with fidelity and trace distance

Quantum channels

To “get inside” the Bloch sphere, we have to apply a different kind of operation: a quantum channel.

A quantum channel Φ maps density matrices to density matrices.

$$\rho \mapsto \rho' = \Phi(\rho)$$

Quantum channels are completely positive, trace-preserving (CPTP) linear maps.

- Trace-preserving: $\text{Tr}(\rho') = \text{Tr}(\rho) = 1$
- Positive: all eigenvalues ≥ 0
- Completely positive: $\mathbb{I}_n \otimes \Phi$ must be positive for all n .

Quantum channels

Channels are characterized by a set of **Kraus operators** $\{K_i\}$,

$$\Phi(\rho) = \sum_i K_i \rho K_i^\dagger$$

$p_i = |\langle \psi_i | \psi \rangle|^2$
 $\rho = \sum_i p_i \rho_i$

where

$$\sum_i K_i^\dagger K_i = I$$

A channel's Kraus operators represent, loosely, a set of things that can happen to a system, including *measurement*, and *errors*.

Example: a channel with one Kraus operator is a unitary channel.

$$\rho \rightarrow \rho' = U \rho U^\dagger$$

$\sum_i K_i^\dagger K_i = U^\dagger U = I$
 $\hookrightarrow K_0$

Example: measurement

Concrete example: $\{M_m\} = \{\Pi_+, \Pi_-\}$:

$$\sum M_m^\dagger M_m = I$$

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

↓ measure,
don't look

$$\frac{1}{2} |0 \times 0\rangle + \frac{1}{2} |1 \times 1\rangle$$

Consider mixed state

$$\rho = \sum p_i \rho_i = \frac{1}{5} |0 \times 0\rangle + \frac{4}{5} |-\times -\rangle$$

Intuitively, after measurement expect

$$\begin{aligned} \rho' &= \frac{1}{5} \left(\frac{1}{2} |+\times +\rangle + \frac{1}{2} |-\times -\rangle \right) + \frac{4}{5} |-\times -\rangle = \frac{1}{10} |+\times +\rangle + \frac{9}{10} |-\times -\rangle \\ &= \sum M_m \rho M_m^\dagger = \Pi_+ \rho \Pi_+^\dagger + \Pi_- \rho \Pi_-^\dagger \end{aligned}$$

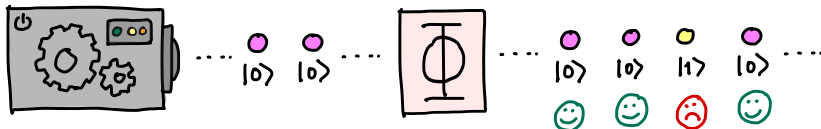
Example: measurement

(Normalized) state after measuring and obtaining outcome m :

Overall state transforms as

The bit flip channel

Suppose a “bit flip” (Pauli X) error occurs with probability p .



How do we write this as a channel? What are the Kraus operators?

$$\Phi(\rho) = \underbrace{(1-p)}_{K_0} \rho + p X \rho X^\dagger$$

$$K_0 = \sqrt{1-p} I \quad \downarrow \quad K_1 = \sqrt{p} X$$

$$= \sum_i K_i \rho K_i^\dagger$$

The bit flip channel

We can visualize the effects of the channel by observing how it deforms the Bloch sphere.

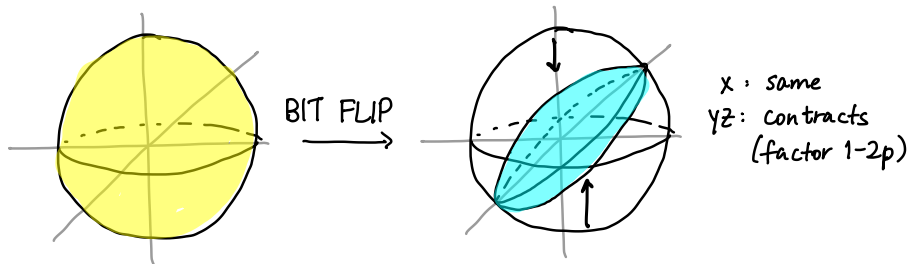
Consider: $\Phi(\rho) = (1-p)\rho + pX\rho X$

- Which states are not affected by a bit flip channel?
- Which states are *most* affected, and where do they go?

Then look at the density matrices of these states.

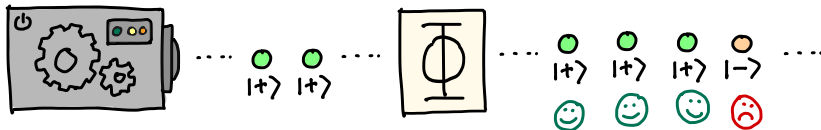
$$\begin{aligned}
 |0\rangle / |1\rangle &: \text{ } \overset{||}{\cap} \\
 X Y \text{ plane} &: \text{ } \overset{||}{\text{---}} \\
 |+\rangle / |-\rangle &: \text{ } \overset{||}{\cup}
 \end{aligned}$$

The bit flip channel



The phase flip channel

Suppose a “phase flip” (Pauli Z) error occurs with probability p .



How do we write this as a channel? What are the Kraus operators?

$$\Phi(\rho) = (1-p)\rho + p \cdot Z \rho Z$$

$$K_0 = \sqrt{1-p} I \quad K_1 = \sqrt{p} Z$$

The phase flip channel

Same questions:

- Which states are not affected by a phase flip channel?
- Which states are *most* affected, and where do they go?

Then look at the density matrices of these states.

$$\rho = \begin{pmatrix} a & b \\ b^* & 1-a \end{pmatrix}$$

$$Z \rho Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a & b \\ b^* & 1-a \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} a & b \\ -b^* & a-1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

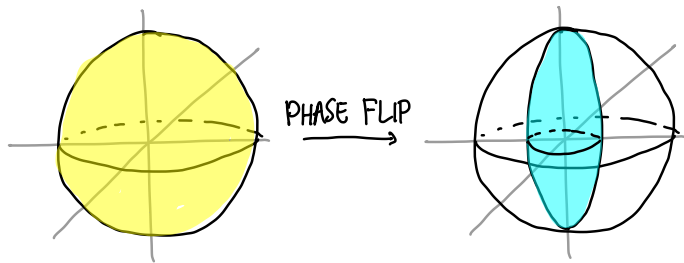
$$= \begin{pmatrix} a & -b \\ -b^* & 1-a \end{pmatrix}$$

$$\rho' = (1-p) \rho + p \cdot Z \rho Z$$

$$= (1-p) \begin{pmatrix} a & b \\ b^* & 1-a \end{pmatrix} + p \begin{pmatrix} a & -b \\ -b^* & 1-a \end{pmatrix}$$

$$= \begin{pmatrix} a & (1-2p)b \\ (1-2p)b^* & 1-a \end{pmatrix}$$

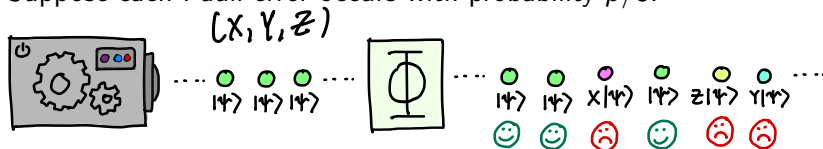
The phase flip channel



z : same
 xy : contracts
(factor $1-2p$)

The depolarizing channel

Suppose each Pauli error occurs with probability $p/3$.



How do we write this as a channel? What are the Kraus operators?

$$\Phi(\rho) = (1-p)\rho + \frac{p}{3}X\rho X + \frac{p}{3}Y\rho Y + \frac{p}{3}Z\rho Z$$

$$K_0 = \sqrt{1-p} I \quad K_1 = \sqrt{\frac{p}{3}} X \quad K_2 = \sqrt{\frac{p}{3}} Y \quad K_3 = \sqrt{\frac{p}{3}} Z$$

Q: What will happen to the Bloch sphere?

The depolarizing channel

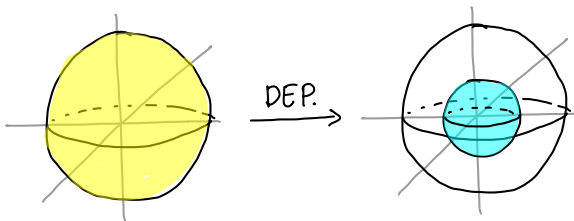
The depolarizing channel

$$\Phi(\rho) = (1-p)\rho + \frac{p}{3}X\rho X + \frac{p}{3}Y\rho Y + \frac{p}{3}Z\rho Z$$

can also be written as

$$\Phi(\rho) = (1-\lambda)\rho + \lambda \cdot \boxed{\frac{I}{2}}$$

max. mixed state.



Amplitude damping channel

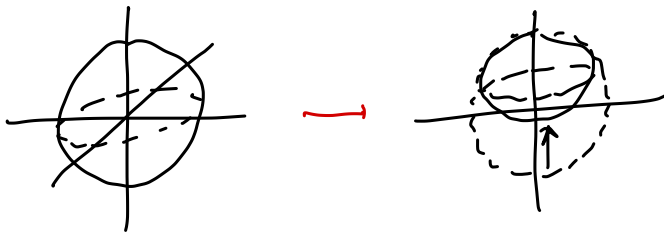
Suppose $|1\rangle$ relaxes to $|0\rangle$ with probability p .

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}$$

$$K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}$$

$\sqrt{p} |0\rangle\langle 1|$

How do you think this affects the Bloch sphere?



Comparing density matrices

Exercise: Suppose we prepare a system in the state

$$|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

If a depolarizing channel with $p = 0.02$ is applied,

- ① What's the probability of observing $|0\rangle$?
- ② How do we quantify the distance from the original state?

Solution 1: solve by hand. Tedious, but can evaluate

$$P_r(0) = \text{Tr}(|0\rangle\langle 0| \cdot \Phi(\rho))$$

↓

$$\Phi(\rho) = (1 - 0.02)\rho + \sqrt{\frac{0.02}{3}} X \rho X + \dots$$

Comparing density matrices

The inner product tells us how close two pure states are:

$$\langle \psi | \phi \rangle$$

close as possible: 1
far as possible: 0

How close are two mixed states ρ and σ ?

One common metric is the **trace distance**:

$$\begin{aligned} T(\rho, \sigma) &= \frac{1}{2} \text{Tr} \left(\sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)} \right) \\ &= \frac{1}{2} \|\rho - \sigma\|_1 \end{aligned}$$

Bounded by $0 \leq T(\rho, \sigma) \leq 1$; lower trace distance is better.

Comparing density matrices

Can also compare using **fidelity**,

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$$

Bounded by $0 \leq F(\rho, \sigma) \leq 1$. *Higher fidelity is better.*

Example: Determine $F(\rho, \sigma)$ if both ρ, σ are pure.

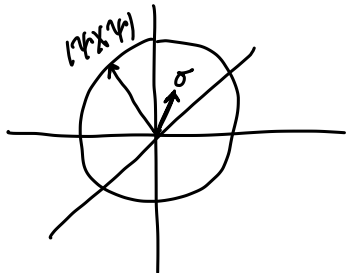
$$\begin{array}{cc} \swarrow & \searrow \\ |\psi\rangle\langle\psi| & |\phi\rangle\langle\phi| \end{array}$$

$$F(\rho, \sigma) = |\langle\psi|\phi\rangle|^2$$

Comparing density matrices

Exercise: Determine $F(\rho, \sigma)$ if ρ is pure but σ is mixed.

$$F(|\psi\rangle\langle\psi|, \sigma) = \langle\psi|\sigma|\psi\rangle$$



$$\sigma = \frac{1}{2} I$$

$$\Rightarrow F(|\psi\rangle\langle\psi|, \frac{1}{2} I) = \frac{1}{2}$$

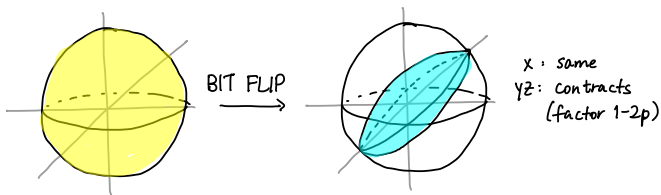
Comparing density matrices

Exercise: What is the fidelity of any pure ρ with the maximally mixed state, $\sigma = \frac{1}{2}I$?

Comparing density matrices

Exercise: Determine $F(\rho, \sigma)$ if ρ is pure and $\sigma = \Phi(\rho)$, where Φ is the *bit flip channel* with parameter p .

left as exercise! (hint: it involves $\langle X \rangle$).



Next time

Next/ class:

- Intro to quantum error correction

Action items:

- ➊ Prepare for quiz 8
- ➋ Bonus assignment (Piazza)
- ➌ Work on project

Recommended reading:

- From this class: Codebook NT, D; N&C 8.2-8.3, 9.1-9.2
- For next class: Codebook E; N&C 10.1-10.2